

Sairaj Nanda

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Education

Bachelor of Science in Data Science and Programming

(Jan 2022 – Dec 2025)

Indian Institute of Technology, Madras

CGPA: 7.9

Work Experience

Data Scientist - Air India Express

(Dec 2025 - Present)

- In-Flight Services (IFS) Revenue & Inventory Management Model: Designed and implemented an end-to-end machine learning pipeline, including data preparation, feature engineering, model development, and deployment. Built predictive ML models to optimize revenue forecasting and improve inventory management, enabling data-driven decision-making and operational efficiency.
- Meal Propensity Model based on Customer trip attributes handled a highly imbalanced dataset.
- Revenue Spoilage Report for daily RM needs and other reporting for tracking to benefit RM need and smoother operation.
- Fare Sync & ODRM updates for Revenue Management team. ODRM Bid price updates and fare sync new product incorporation.
- End to end data pipeline for regular reporting and automation.
- Tableau & Power Bi Dashboard updates and Creations.

Research Intern – IIT Madras

(Jan 2023 – Jul 2023)

- Developed and fine-tuned NLP and Speech Recognition models including ESPNet, HMM, and Hybrid Segmentation for 13 Indian languages, enhancing multilingual parsing accuracy.
- Integrated Gmail API into project infrastructure to automate dataset communication and user notifications, improving operational efficiency.

Projects

RAG Architecture

[[GitHub](#)]

- Implemented a Retrieval-Augmented Generation (RAG) pipeline using Flask, LangChain, and Workers AI API for efficient document-based question answering.

AI Study Assistant Portal

[[GitHub](#)]

- Built an AI-powered student learning platform replicating the SEEK portal using React and Flask.
- Integrated RAG chatbot to assist students in interactive learning, improving user engagement by 30%.

Food Rating Predictor

[[GitHub](#)]

- Designed ML models to predict food ratings based on text reviews using vectorization and sentiment analysis.
- Leveraged Stacking Classifier and Linear SVM to boost predictive accuracy on imbalanced data by 12%.

Sponsor-Influencer Engagement Platform (v2)

[[GitHub](#)]

- Enhanced V1 using Vue.js, Redis, Celery, and Mailhog for caching, scheduling, and email automation—reducing task latency by 25%.

House Website

[[Live Site](#)]

- Designed a community website enabling students to track events, verify membership, and register seamlessly. Improved onboarding efficiency by 40%.

Next.js Projects – Election & Fest Websites

- Developed secure voting and fest registration systems with automated email confirmations and Firebase integration.

Object Detection Model

[[GitHub](#)]

- Built real-time object detection using YOLOv5 in Python to identify multiple object classes with high accuracy.

Skills

Programming Languages

Python, SQL, HTML, C++

Frameworks & Tools

Flask, REST API, Vue CLI, Jinja2, Bootstrap, Redis, Flask-SQLAlchemy, Git, Bash, Node.js, Next.js, React, Streamlit, Django, Pandas, Numpy, PostgreSQL, Postman, Tableau, Power bi

ML/DL Libraries & Techniques

Scikit-learn, NLP, Speech Technologies, Tesseract, Embeddings, Matplotlib, Seaborn